



WELCOME





Exploring Gen AI Capabilities



Think of a task or problem
you faced recently where you think
Gen AI could have been helpful



Let's explore the most practical
and powerful capabilities
we can start using immediately



Agenda for Today's Session

Exploring Gen AI Capabilities



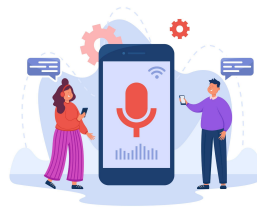
Search

Getting real-time,
accurate information
faster than ever before



Deep Research

Conducting
comprehensive,
multi-source research in
minutes instead of hours



Voice Interaction

Having natural
conversations through
your voice



Comparing Different LLMs

Understanding when to
use which models

Search (Web Browsing)





Search (Web Browsing)

Modern LLMs like **ChatGPT**, **Claude**, and **Gemini** can search for recent or real-time information on the internet





Search (Web Browsing)

Helpful for questions about current events, unfamiliar topics, or when you want source-backed responses



**Current
Events**



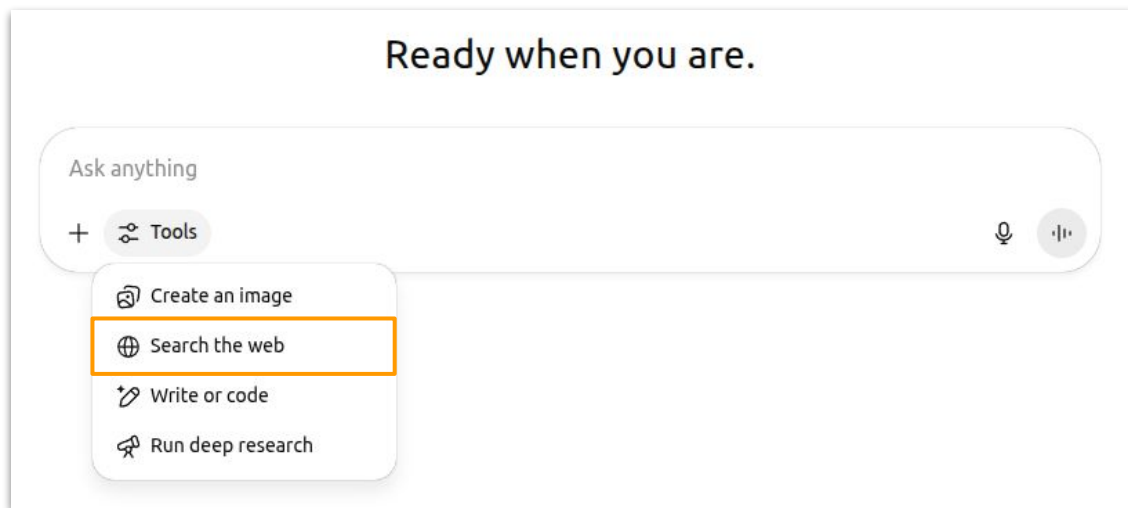
**Unfamiliar
Topics**

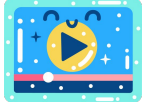


**Source-backed
Response**

Gen AI Capabilities - Deep Research

Search - ChatGPT





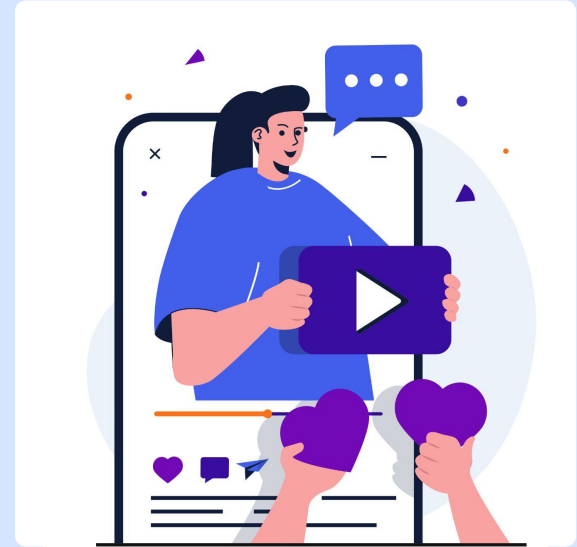
Trending Reels



Type your text here

| Generate

What are the trending Instagram reel challenges right now?





Local Recommendations



Type your text here

| Generate

Show me the best restaurants near me





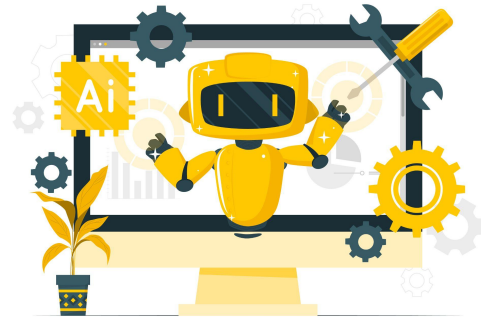
Top AI Tools



Type your text here

| Generate

What are the top 3 AI tools launched
this week?





Financial Updates



Type your text here

| Generate

How has Apple's stock performed over the last week?





Search (Web Browsing)

The LLM response will contain inline citations, and we can **hover** over the **citations to learn more**



Search

Using Across Domains



Agriculture

To find crop care tips, pest control methods, and weather forecasts



Sports

For live scores, player stats, and upcoming match schedules



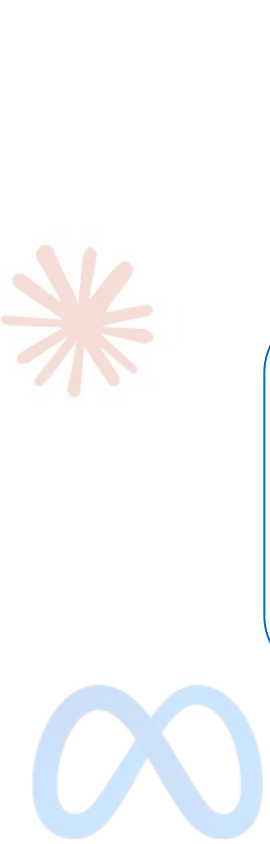
Education

For the latest course materials or tutorial videos on specific subjects



Fashion

For the latest trends, and brand comparisons in online stores



**Think about use cases where
you can use the search feature**



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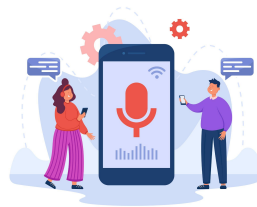
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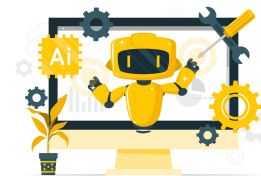
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Understanding when to
use which models



Deep Research

Sometimes a **quick search** isn't enough





Deep Research

For complex decisions or in-depth understanding, we **need comprehensive research** that **explores multiple sources** and presents a complete picture





Deep Research

Designed to perform in-depth, **multi-step**
research using data from the web



Search



Extract



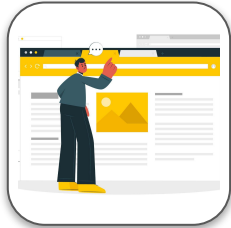
Analyze



Summarize



Deep Research as a Team of Researchers



Reading Through
Webpages



Analyzing Information



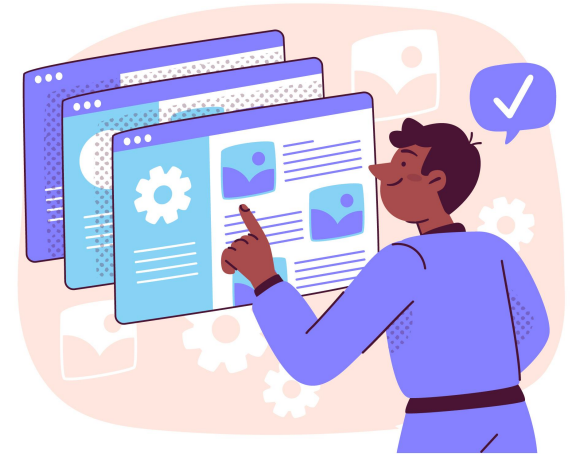
Comparing Different
Viewpoints





Deep Research

It is great for people who need to **find detailed and accurate information** from various websites





Let's say you want to buy a laptop

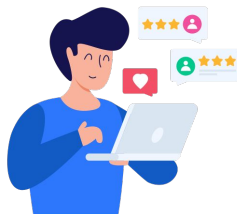


Buying a Laptop

Instead of spending hours comparing specs across:



Different Websites



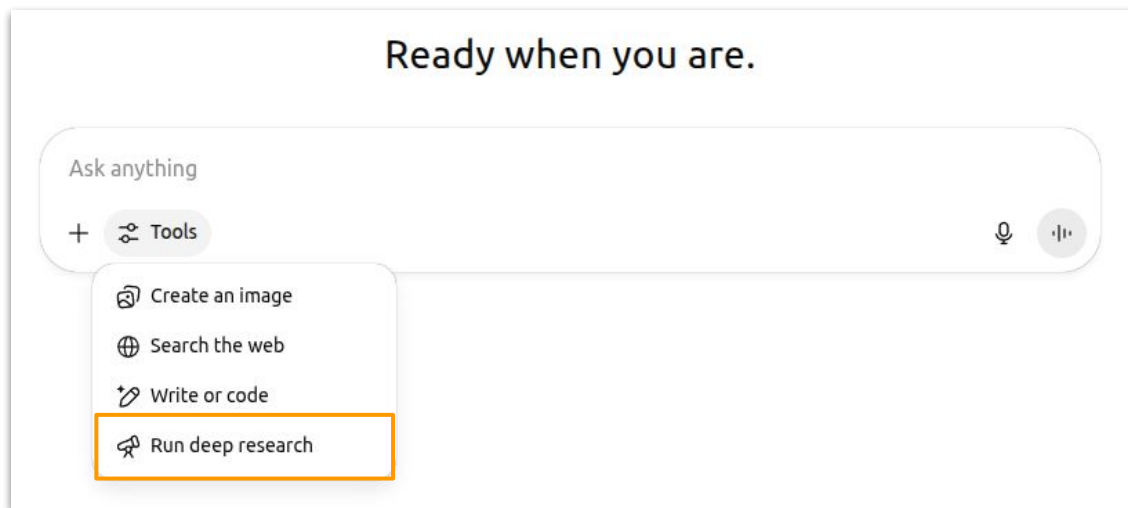
Reading Reviews



Checking Prices

We can use deep research feature

Deep Research - Chat GPT





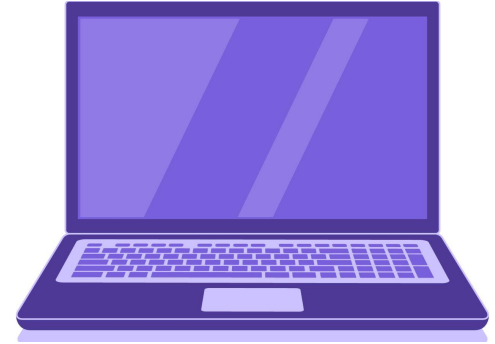
Deep Research for Purchasing



Type your text here

| Generate

I am planning to buy a new laptop and need a detailed comparison to make the right decision. My budget is ₹60,000 to ₹80,000, and I will mainly use it for web development and occasional video editing. Please suggest 3-5 laptops that are best suited for this use case





Deep Research for Career Planning



Type your text here

| Generate

I am a computer science student interested in AI/ML careers. Can you do deep research on:

1. The current job market for AI engineers in India
2. Skills and certifications most in demand
3. Average starting salaries
4. Growth prospects over the next 5 years
5. Companies hiring the most AI engineers in India





Deep Research

Deep research typically **completes a query within 5 to 30 minutes**, depending on its complexity



Quick Note

NOTE

There will be a daily limit for using the
search and deep research features

Try using Claude, Gemini, and Llama



Use Cases for Deep Research



Fashion

Health Care

Education

Can you think of use cases where
deep research can be used in
different industries?

Finance

Agriculture

Social Media



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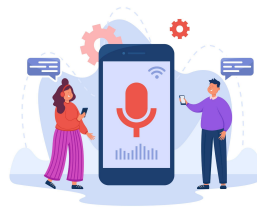
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Types of AI Models

Understanding when to
use which models

Voice Input and Output





Voice Input

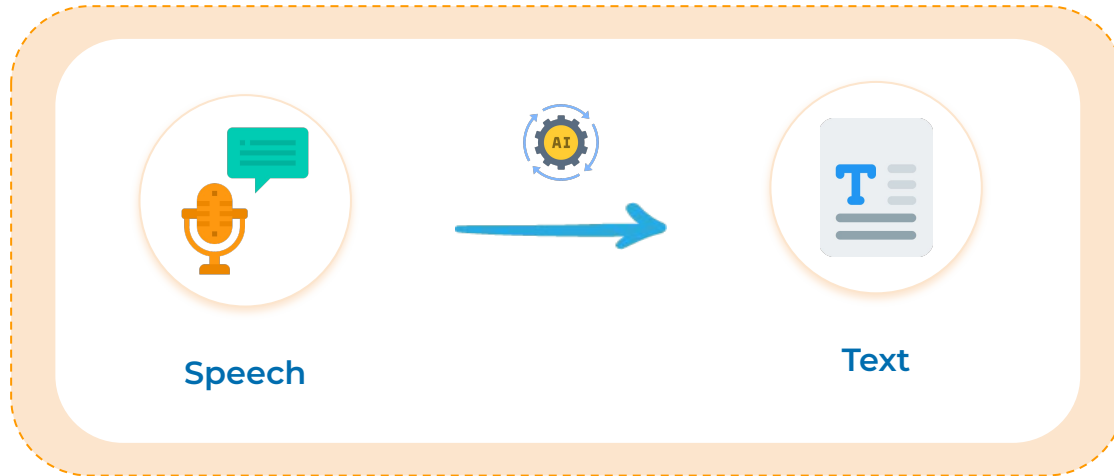
Modern LLMs can **understand natural speech** and **respond with human-like voices**, making interactions more intuitive and accessible





Voice Input

They can accept voice input, allowing us to speak directly, with our **speech being converted into text for processing**





Voice Input - Voice to Text

What's on your mind today?

Ask anything

+ Tools



Type your text here

| Generate

Hey, Can you suggest the best places to visit in India?



Quick Note

NOTE

There will be a daily limit for using
the deep research and
advanced voice mode features

Try using Claude, Gemini, and Llama



Voice Input - Advanced Voice Mode

What's on your mind today?

Ask anything

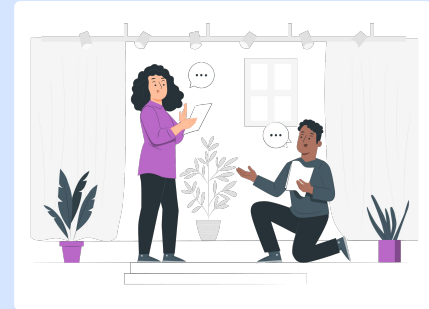
+ Tools



Type your text here

| Generate

Act as my best friend who cares about me a lot.
Let's role-play a conversation





Practice Speaking and Listening



AI Assistant



Type your text here

Generate

Can you teach me English by having a conversation with me?
Please ask me questions in between and correct my mistakes



Speaking Practice



AI Assistant



Type your text here

Generate

I need to practice my presentation speech for tomorrow. Can you listen and give me feedback? My topic is Generative AI



Preparing for Interview(mock interview)



AI Assistant



Type your text here

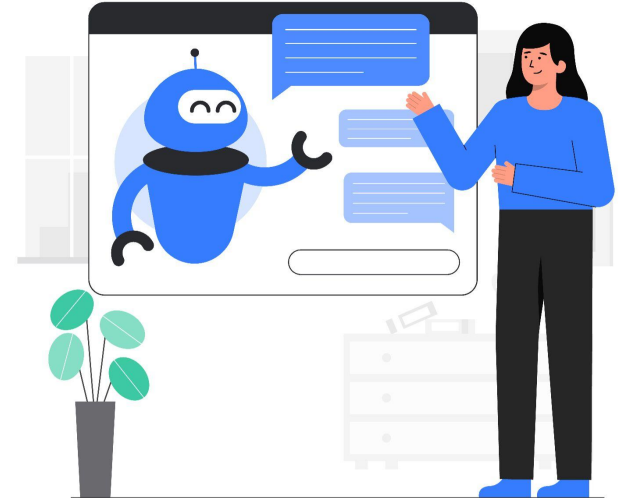
Generate

Hey, I want to prepare for an AI Engineer fresher
Role? Can we have a mock interview?



Chat GPT Voice Mode

- ✓ During a voice chat, we can **ask ChatGPT to search** for something and it **will provide us with fast, timely answers**





**In what use cases would voice
interaction be most helpful?**



Screen Sharing Capability





Screen Sharing Capability

We can also share the screen and ask questions. This feature **allows LLMs to see what's on your screen**





Use Cases



Code Debugging
and Review



Filling Complex
Forms



Resume/
Document Review



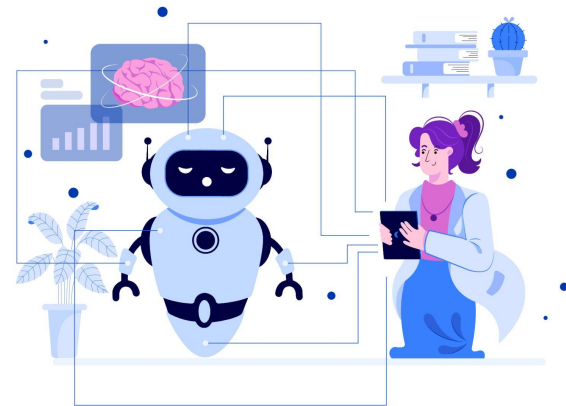
Screen Sharing Capability

Screen sharing is currently in the
**experimental phase within ChatGPT's
Advanced Voice Mode**





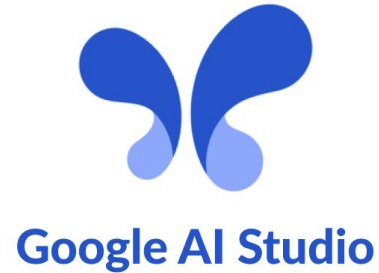
Let's use the Google AI Studio's
screen share feature





Screen Sharing Capability

Google AI Studio is a platform where we can access and experiment with Google's AI models, including Gemini





Example

Let's say you are creating or editing a PPT for your project presentation, and you **don't know how to insert an image**





While Creating PPT



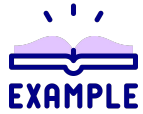
Google AI Studio



Type your text here

Generate

Hey, I am working on this slide. Can you guide me with the steps on on how to insert an image on it?



Debugging Code



Google AI Studio



Type your text here

Generate

I am getting an error in my Python code. Can you look at my screen
and help me figure out what's wrong?



Applying to Job



Google AI Studio

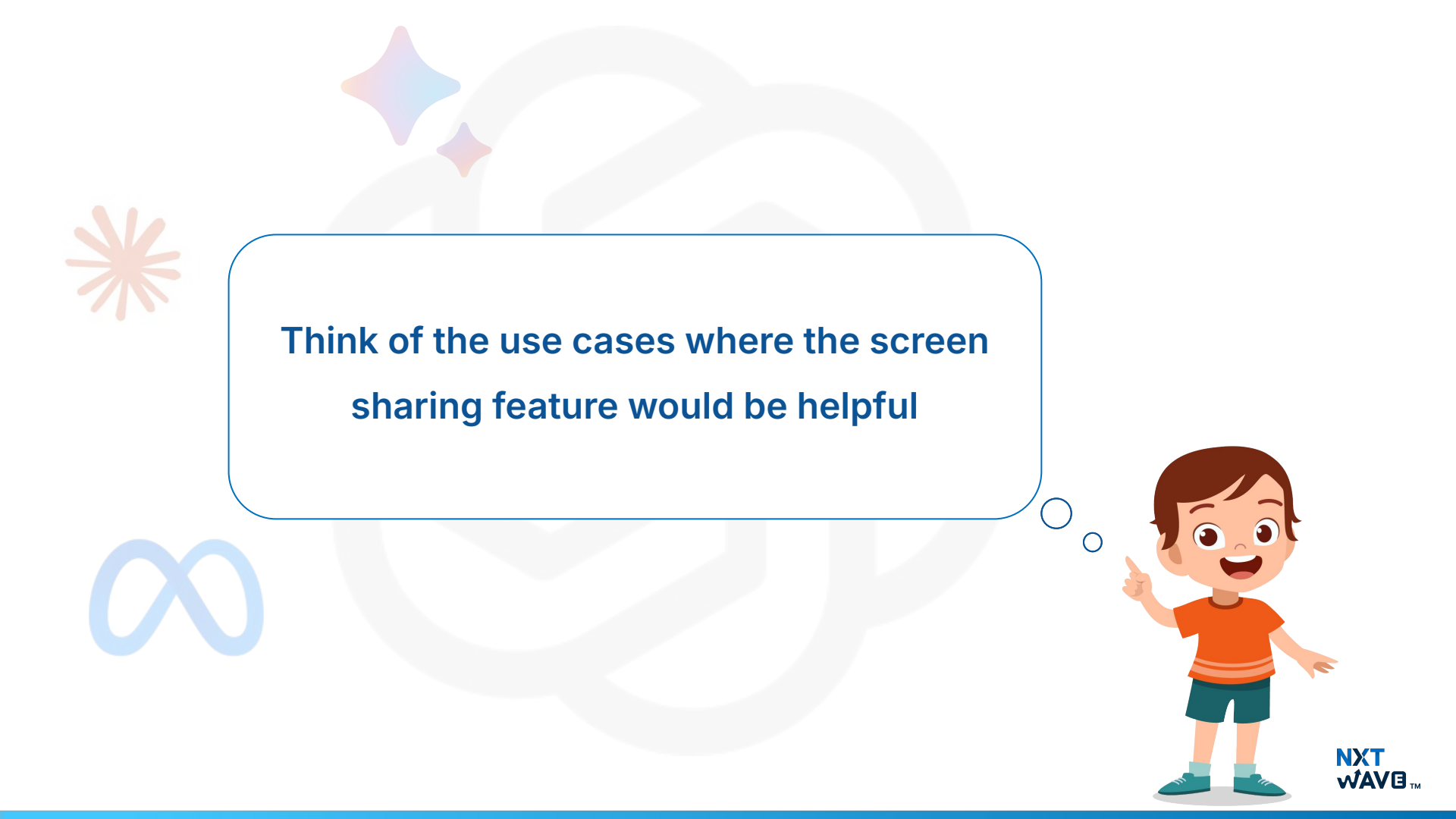


Type your text here

Generate

What does 'Stream' mean here? Should I select 'IT' or 'Engineering'?

I want to apply for the AI Engineer role, and I am from CSE



Think of the use cases where the screen sharing feature would be helpful



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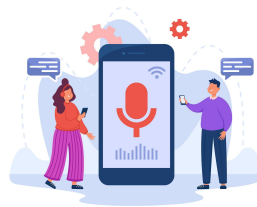
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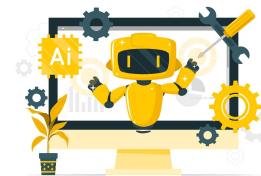
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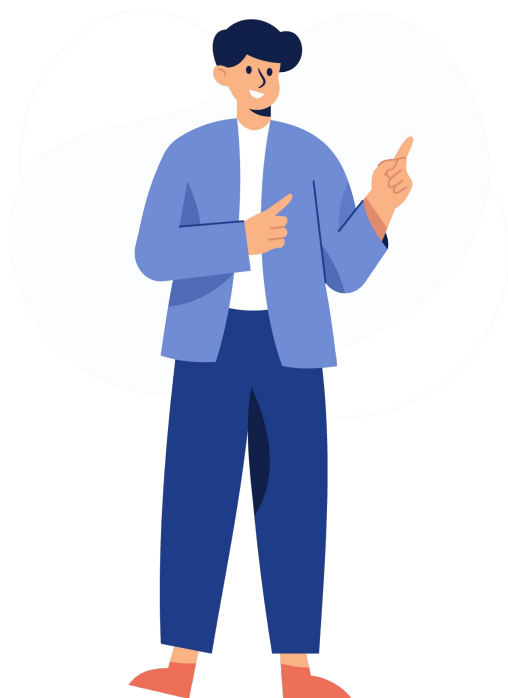


Understanding Different AI Models

Just as we would not use a hammer for every task in a toolbox, **different types of AI models are optimized for different kinds of tasks**



Most Common Model Categories



Reasoning Models



Flash Models



Coding Specialists



General Purpose



Cost-Efficient



Small Language Models



Reasoning Models

Reasoning models are helpful
in **handling complex logical tasks**
through **step-by-step thinking processes**





Reasoning Models



Breaking down multi-step problems into smaller components



Following logical chains of thought



Supporting deeper analysis and more thorough consideration of problems

Top Reasoning Models



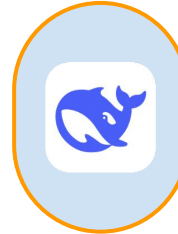
o3



Claude Opus 4



Gemini 2.5 Pro



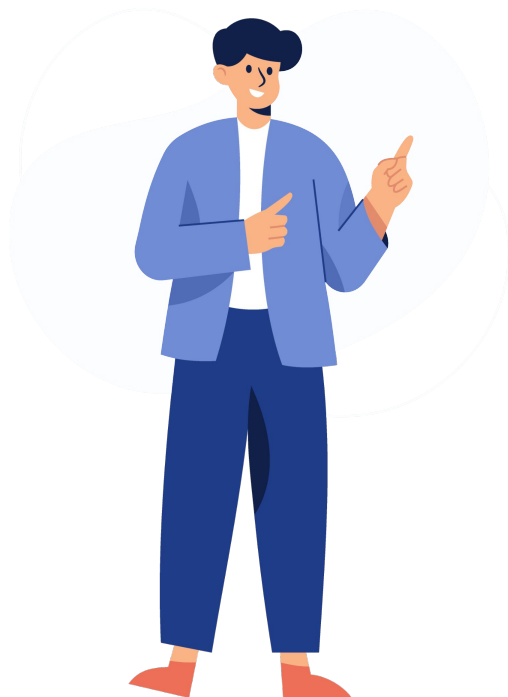
DeepSeek R1

Example

What is the next number in the series?

2, 6, 12, 20, 30, ___?

Most Common Model Categories



Reasoning Models



Flash Models



Coding Specialists



General Purpose



Cost-Efficient

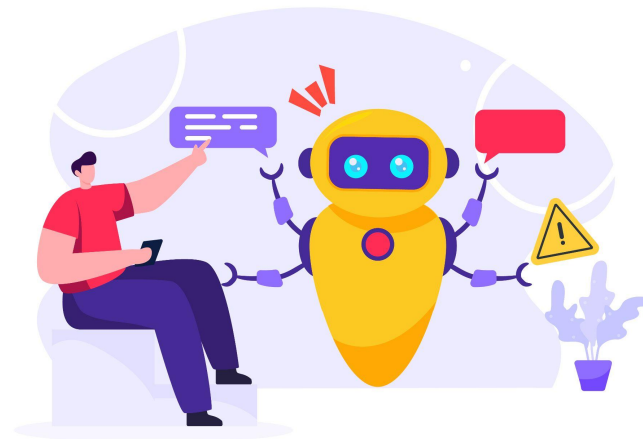


Small Language Models



Flash Models

Flash models (sometimes called "fast" models) are **optimized for speed and efficiency, rather than deep reasoning**





Flash Models



Quick responses to straightforward queries



Efficient processing of routine tasks



Lower computational resource requirements



Handling high volumes of interactions

Most Common Model Categories

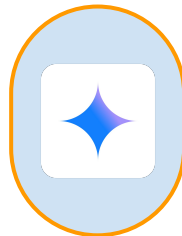
Top Flash Models



OpenAI
GPT-4.1 nano



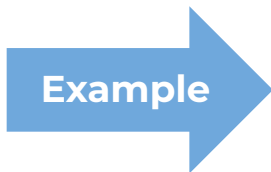
Anthropic
Claude 3.5 Haiku



Gemini 2.5 Flash-Lite



DeepSeek-V3



Create a simple weekly workout plan
for a beginner

Models

Specialized in Coding

- Anthropic Claude Opus 4
- OpenAI GPT-4.1
- DeepSeek-Coder-V2
- Meta Code Llama

General Purpose: For day-to-day Use

- GPT-4o,
- GPT-4o-mini
- Claude Sonnet 4
- Gemini 2.5 Flash
- Llama 4 Maverick

Cost Efficient

- DeepSeek(all models)
- Llama 4
- Gemma 3n
- Phi-3-mini

Choosing an LLM

Make sure you choose the **'best'** LLM depending on your needs..!



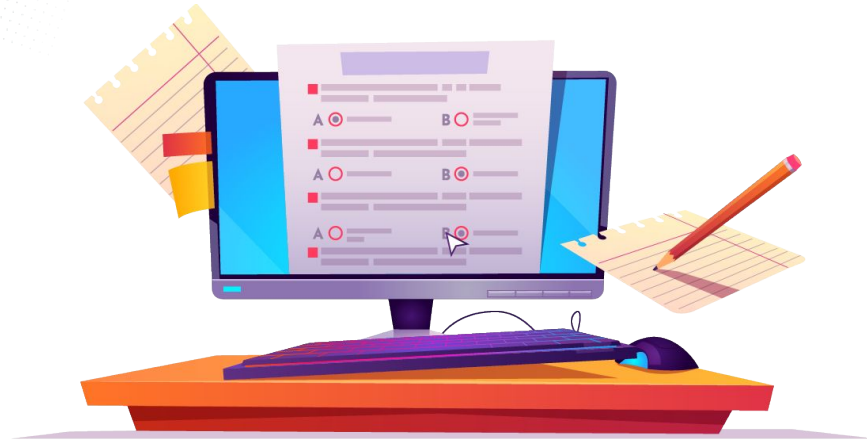
Exploring Gen AI Capabilities

Key Takeaways

- ✓ Search (Web Browsing)
- ✓ Deep Research
- ✓ Voice Input and Output
- ✓ Understanding Different AI Models



Practice



Feedback





THANK YOU





ALL THE BEST

